

ECOGRAPHY

Research Article

- 0 DNA barcoding of more than one million insect specimens 61
indicates that at least one third of California's insect biodiversity
Q1 remains undiscovered 65
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25 Insects are the most diverse terrestrial organisms, yet they are underrepresented in large-scale conservation assessments. To
address this gap, the California Insect Barcoding Initiative is developing a publicly accessible, statewide DNA-barcode refer-
ence library for California insects to support scalable surveys, establish baseline biodiversity measurements, and generate
30 potential distribution maps that inform conservation planning. Specimens are collected with a hybrid strategy that combines
standardized Malaise trapping, to enable replicable sampling, and opportunistic collecting, to maximize taxonomic coverage.
Using samples collected primarily from 2022–2024 at 72 longer-term Malaise trap sites, coupled with collecting using varied
35 methods at 454 additional locations across California, we have barcoded over one million specimens. Preliminary complete-
ness analyses suggest that current sampling captures roughly 65–69% of the fauna, implying a conservative minimum of circa
61 000 insect species in California, as delimited by their DNA barcode sequence. Using all sequenced specimen records, we
40 generated rule-based spatial range interpolations constrained by ecoregion and vegetation type, and used these to infer spatial
patterns in insect species richness across the state. We identify areas of both high and low potential species richness, with cur-
rent peaks in the Southern California Mountains ecoregion and the Mojave Basin and Range ecoregion. Our species richness
estimates and spatial patterns are explicitly provisional and are expected to evolve as sampling gaps are addressed. Finally, we
45 make all sequence data, specimen images, and occurrence records publicly available via the Barcode of Life Datasystem. This
ongoing effort constitutes the first large-scale DNA barcode-based survey for California insects, providing an expandable foun-
dation for tracking temporal change, testing drivers of insect diversity, and prioritizing regions for conservation and targeted
inventory.

Keywords: biodiversity, California, DNA barcoding, Insecta

Q3 Introduction

50 Our planet is losing species at an unprecedented rate, which many scientists suggest may be an anthropogenic mass extinc-
tion event (Cowie et al. 2022). This includes losses in one of
the most diverse and abundant groups of organisms on the
55 planet – the insects (Wagner et al. 2021a). In most terrestrial
ecosystems insects represent the largest proportion of ani-
mal diversity. Early estimates of insect biodiversity suggested
that there are approximately 30 million insect species world-
60 wide (Erwin 1983), while more recent estimates converge
around 6 million species (Novotny et al. 2002, Basset et al.
2012, Stork et al. 2015). New cryptic species revealed by
DNA sequencing further increase these estimates to circa
20–40 million insect species worldwide (Larsen et al. 2017,
Li and Wiens 2023). Insects’ importance for maintaining
115 healthy food webs and ecosystem services (e.g. pollination,
pest control, dung burial) was valued at \$57 billion annu-
ally in the USA in 2006 (Losey and Vaughan 2006). When
adjusted for inflation, this figure is equivalent to \$109.1 bil-
lion in 2025 (Bureau of Labor Statistics n.d.). Insects’ tremen-
dous diversity and provisioning of critical ecosystem services

0 make their ongoing, severe population declines extremely
troubling (Shortall et al. 2009, Hallmann et al. 2017, Janzen
and Hallwachs 2021, Dalton et al. 2023).

5 Despite the ecological importance of insects and evidence
of their global decline, critical data for insect distributions and
population trends are lacking for most areas. This is problem-
atic for conservation efforts, as distribution maps are essen-
tial for spatial conservation planning (Porfirio et al. 2014).
10 For instance, conservation practitioners in California use the
ACE 3.0 Terrestrial Biodiversity Summary of California maps
to guide decision making (California Department of Fish and
Wildlife 2018); however, these maps only incorporate data
from birds, mammals, amphibians, reptiles, and plants, which
15 may serve as poor proxies for insects and other invertebrates
(Oberprieler et al. 2019). Moreover, to date most research on
insect distributions, diversity, and sampling completeness has
focused on a limited subset of lineages, primarily charismatic
groups like bees (Cheshire et al. 2023), butterflies (Sánchez-
20 Fernández et al. 2021, Shirey et al. 2021), and moths (Janzen
and Hallwachs 2021, Wagner et al. 2021b), leaving many
major taxonomic groups a mystery. Addressing these data
gaps requires large-scale insect surveys, development of spe-
cies checklists, and methods for documenting and delimiting
25 unknown species.

Understanding insect diversity patterns is valuable not only
for their conservation, but also because several aspects of their
biology make insects an excellent model system for monitor-
ing broader, long-term changes in community biodiversity.
30 For instance, many insects are highly endemic and sensitive
to changing environments (Parmesan et al. 1999), and many
species can serve as indicators of shifts in local seasonality,
given their dependence on temperature for development (Roy
and Sparks 2000, Forrest 2016). Moreover, insects exhibit
35 higher total species richness and abundance than vertebrate
groups (Mora et al. 2011, Rosenberg et al. 2023), facilitating
more precise comparisons of diversity across sites and over
time. Insects are also relatively easy to passively sample com-
pared to vertebrates, which often require capture-and-release
40 or field observations at the right time and place (Sutherland
2006). Consequently, insect biodiversity is already used as a
staple of assessing the health of freshwater ecosystems (e.g.
the Ephemeroptera, Plecoptera and Trichoptera (EPT) Index
(Blachuta et al. 2014)).

45 Despite these advantageous characteristics, insects are
rarely, if ever, used in terrestrial biodiversity estimates for con-
servation planning (Nania et al. 2024). Moreover, significant
monitoring gaps exist for insects, largely because identifying
50 large numbers of specimens, collected or observed, to the
species level is difficult, time-intensive, and can require high
levels of taxonomic expertise. Only a few efforts are being
made in the United States to perform long-term insect popu-
lation monitoring at scale, including the National Ecological
55 Observatory Network (NEON; (Thorpe et al. 2016)) and the
Long-Term Ecological Research (LTER; Crossley et al. 2020)
sites. While these programs represent unique opportunities
to monitor insect biodiversity across the country, they have
60 focused on specific groups of insects, largely determined by

61 collection methods used (e.g. Coleoptera using pitfall traps or
mosquitoes using carbon dioxide traps) (Crossley et al. 2020,
Thibault et al. 2023). More systematic approaches for moni-
toring insects are needed to improve the efficiency (in terms
65 of cost and quantity) of both collecting and identifying vast
amounts of insect specimens.

DNA barcoding is a rapid and affordable approach that
has become widely used in the past several decades for iden-
tifying and even delimiting species based on the sequences of
70 short segments of DNA (e.g. the cytochrome c oxidase sub-
unit I (COI) locus for animals; (Hebert et al. 2003, Wilson
2012). DNA barcoding approaches have enabled insect bio-
diversity studies to extend beyond groups that are readily
identifiable from morphology alone (e.g. butterflies, bumble-
75 bees) to include groups that are highly diverse but largely
undescribed (often referred to as ‘dark taxa’, e.g. many small
flies and wasps) (Meier et al. 2025). DNA-based observato-
ries are increasingly recognized as essential infrastructure for
scalable biodiversity monitoring and forecasting, integrating
80 barcode data with ecological models and global survey net-
works (Ovaskainen et al. 2025).

Several major regional efforts have developed large-scale
insect inventories using DNA barcoding. In the Área de
85 Conservación Guanacaste (ACG) in northwestern Costa
Rica, taxonomists have undertaken insect sampling since
1978 and implemented voucher-based DNA barcoding start-
ing in 2003 (Janzen and Hallwachs 2011). This effort formed
the basis for the All Taxa Biodiversity Inventory (ATBI) con-
cept (Janzen and Hallwachs 1994). The ACG dataset pri-
marily consists of reared caterpillars and their parasitoids,
providing valuable data on multi-trophic interactions and
95 on a highly species-rich dark taxon, parasitoid insects. The
combination of these reared samples with an extensive series
of Malaise traps allows the two data sources to inform each
other, especially for understanding completeness of the ATBI
efforts (Quicke et al. 2024, Colwell et al. 2026). A survey in
the Southern Appalachian Sky Islands leveraged DNA bar-
100 coding of specimens from leaf litter samples to investigate
community composition and local endemism (Caterino
and Recuero 2024). In Canada, a multi-method approach
has been used to DNA barcode one million insect specimens
from across the country, with the ultimate goal of develop-
105 ing more accurate planetary species richness estimations
(Hebert et al. 2016). Many areas around the world would
benefit from similar sampling efforts, and a key priority for
the research and conservation communities should be to
extend surveying of insects in areas identified as biodiversity
110 hotspots (Myers et al. 2000).

The California Floristic Province is one such hotspot. It is
the most biodiverse region in North America (Calsbeek et al.
2003), and is considered to be in the world’s top 25 most
115 biologically rich and threatened terrestrial ecosystems
(Myers et al. 2000). Roughly 87% of this biodiversity hotspot
occurs within the state of California, and it covers approxi-
mately 70% of the state’s land area.

The California Floristic Province contains ~ 20% of all
120 vascular plant species known from the United States and

0 more than 30% of known insect species north of Mexico
 (Calsbeek et al. 2003, Franklin et al. 2021). Though the state
 of California harbors substantial insect biodiversity, most
 insect taxa lack a comprehensive checklist of species present
 5 in the state, a data gap that is particularly acute for the most
 diverse groups with high numbers of dark taxa (e.g. Diptera,
 parasitic Hymenoptera). As a result, the true proportion of
 North American insect species found in California remains
 unknown, hampering conservation efforts.

10 Efforts to understand and conserve California's biodiversi-
 ty require more complete knowledge of the state's insect
 fauna. In alignment with the 2030 targets of the Kunming-
 Montreal Global Biodiversity Framework (Convention on
 Biological Diversity 2022), California has committed to
 15 preserving 30% of land within the state by 2030 (California
 Natural Resource Agency 2021). Achieving these targets
 requires identifying the distribution of available conservable
 land in California, coupled with clarifying the spatial distri-
 bution of biodiversity in the state. Without insect distribu-
 20 tion maps, we lack critical information needed to incorporate
 these important species within this framework. Moreover,
 detecting the effects of climate change and biodiversity loss
 requires monitoring to begin immediately to establish a base-
 line and continue into the future.

25 To facilitate these goals, natural history museums, research
 institutions, and other partners across California have come
 together to collect and sequence DNA barcodes for every
 insect species present in the state. This state-funded collabora-
 30 tive project, called the California Insect Barcoding Initiative,
 is part of a larger all-taxon biodiversity inventory effort in
 California, CalATBI. The project's primary objectives are to
 establish a voucher-based DNA barcode reference library for
 California insects to improve the efficiency of future moni-
 35 toring, provide a biodiversity baseline for future sampling
 comparisons, generate potential range maps for insect bio-
 diversity that can inform conservation metrics, and preserve
 newly-collected specimens for future study, while rescuing at-
 risk specimen collections from degradation or disposal.

40 This project's methodology was designed to maximize
 the biodiversity sampled across space and time, generat-
 ing a dataset that can be used to catalog species, discover
 new taxa, track phenology, strategically target undersam-
 45 pled areas, and explore patterns of insect biodiversity.
 Development of a DNA barcode reference library requires
 sampling authoritatively identified specimens from legacy
 collections, coupled with making new collections in the
 field to capture species that are unknown or unrepresented
 50 in collections. The present study focuses solely on the new
 field collection data, prior to the integration of identified
 specimens from legacy collections; therefore, we rely on
 Barcode Index Numbers (BINs; Ratnasingham and Hebert
 55 2013) as a proxy for species. BINs are persistent operational
 taxonomic units generated by the Refined Single Linkage
 (RESL) algorithm on BOLD. The goals of the present study
 are to:

- 1) assess statewide species richness and sampling completeness across the project using Chao species richness estimators; 61
- 2) infer statewide patterns of insect community dissimilarity; 65
- 3) use rule-based spatial range interpolations to generate potential range maps for all insect species collected in California, and
- 4) use these predictions to infer spatial patterns in potential insect biodiversity across the state. 70

This dataset represents one of the largest and most diverse insect barcoding datasets to date, with 1 030 728 sequenced specimens belonging to 42 305 unique BINs. This dataset facilitates the assessment of current sampling completeness and preliminary assessments of broadscale insect biodiversity patterns across California. The California Insect Barcoding Initiative is laying the groundwork for future insect surveys and monitoring programs by developing a DNA barcode reference library to enable rapid identifications, developing 80 a methodology that can be replicated to track phenology or serve as a model for other insect biodiversity surveys, and developing distribution and biodiversity mapping protocols that support targeted resampling and conservation planning. 85

Material and methods

Data collection overview

90 One of the primary goals of the California Insect Barcoding Initiative was to create a comprehensive reference library of DNA barcodes for all hexapod species (insects, collembolans, proturans, and diplurans; henceforth referred to as insects) of California, including both known species and those new to science. To facilitate this goal, we maximized both the spatial and temporal coverage of our field collecting by sampling across as many ecoregions as possible and by revisiting locations throughout the year. We used a two-pronged approach based on 1) Observatory Sites, locations where standardized collecting was conducted by running Malaise traps continuously, and 2) Snapshot Sites, locations sampled at single time points using a wide variety of collecting methods. Snapshot Sites were selected to cover as many different ecoregions as possible, maximizing coverage of distinct plant communities and abiotic factors (Environmental Protection Agency Level IV Ecoregions (Griffith et al. 2016)). Most samples were collected fresh for this project starting in 2022, aside from a small number of samples that were retrieved from cold storage dating back to 2016. Achieving this large-scale sampling effort required a multi-institutional collaborative approach, with participation from several natural history museums, universities, government agencies, and other groups (Supporting information). 115

Observatory Sites

Observatory Site collections (Supporting information) used Townes-style Malaise traps (Townes 1972) purchased from 121

0 Sante Traps (Lexington, Kentucky). Traps typically operated
for one-week intervals followed by sample pickup, unless cir-
cumstances like collector unavailability rendered weekly sam-
pling infeasible. Insects were collected into 95% ethanol, and
5 samples were stored in a cool dry place (freezers or refrigera-
tors when available) until batches of samples were retrieved
by California Insect Barcoding Initiative field crews for stor-
age in freezers.

10 The standardized approach used at Observatory Sites
generated abundance data in addition to presence data,
producing baseline estimates of both species diversity and
abundance to facilitate future efforts to monitor insect com-
munities. Many Malaise traps were deployed near weather
stations, allowing factors like precipitation, temperature, and
15 humidity to be correlated with insect phenology.

For Observatory Site samples, we recorded samples' wet
weight (Supporting information) following the protocols
used by the Centre for Biodiversity Genomics (University
of Guelph, Guelph, Ontario, Canada). After weighing,
20 Observatory samples were sequenced in full with only excess
specimens of ant (Formicidae), honeybee *Apis mellifera*, and
Collembola morphospecies removed to reduce sequencing
redundancy for groups that will not yield additional spe-
cies. Since very few species are affected by this sorting, it will
25 have little to no impact on abundance measurements used
in diversity estimations. All other groups, including those
that we expect to represent dark taxa with high potential
for species discovery, were included in full for sequencing.
30 For every Observatory Site location, one sample per month
was submitted for sequencing at the Centre for Biodiversity
Genomics. Additional Observatory Site samples which were
not immediately sent for sequencing are being stored in freez-
ers at the collection institutions.

35 In total, 72 Observatory Site locations are included in the
dataset, representing 13 of 14 Level III EPA ecoregions in
Q5 California (Fig. 2A). For clarification of terms, samples refer
to bulk samples containing many specimens from a single
collecting event, and specimens refer to the individual insects.
40 Collecting at Observatory Sites resulted in between 2 and 102
bulk samples per site. A total of 956 Observatory Site sam-
ples were collected, from which 536 samples were selected for
sequencing (Supporting information). For each Observatory
45 Site, between 1 and 22 (mean = 7) bulk samples were fully
sequenced at the Centre for Biodiversity Genomics, yielding
a total of 840 260 extracted and vouchered specimens.

50 Snapshot Sites

Snapshot Site collecting approaches were not standardized,
as collectors needed the flexibility to collect for variable dura-
tions as deemed necessary, with methods optimized for dif-
ferent environments and for particular target insect taxa.
55 Methods comprised sweeping and beating vegetation, hand
collecting, short-term Malaise trapping, pan trapping, blue
vane trapping, pitfall trapping, blacklight trapping, mercury
vapor light trapping, sifting leaf litter, dip netting, Lindgren
funnel trapping, and Berlese and Winkler extractions of soil
60 and leaf litter.

61 Insects were generally collected directly into 95% etha-
nol (or transferred into 95% ethanol from soapy water in the
case of pan traps) and were then stored in a freezer prior to
sequencing. In total, 1280 Snapshot sampling events were
65 spatially clustered, such that all sampling events within 1 km
of each other are considered a single site, resulting in 454
Snapshot Sites in this dataset. These sites represent collections
from all Level III EPA ecoregions in California. We collected
at Snapshot Sites using a variety of collecting methods, with
70 many sites revisited for collecting multiple times.

Snapshot Site samples were not weighed, and all taxonomic
groups were pre-sorted prior to sequencing by removing
duplicate specimens within morphospecies, so as to reduce
sequencing redundancy and to voucher additional specimens
75 in institutional collections. All sorted samples were then sent
to the Centre for Biodiversity Genomics for sequencing fol-
lowing the same protocols for Observatory Site samples.

We sequenced 1280 sorted samples derived from Snapshot
80 Site collecting (Supporting information), yielding a total of
190 468 extracted and vouchered specimens.

Specimen preparation and imaging

85 At the Centre for Biodiversity Genomics, specimens were
size-sorted. Individuals smaller than 5 millimeters were
arrayed directly into 96-well microplates, while specimens
larger than 5 millimeters were either pinned, placed in a
matrix tube, placed in a scintillation vial, or placed in an
envelope as appropriate for their taxonomy and then arrayed
90 in a 96-grid Schmidt box. High resolution images were taken
of each specimen, with specimens in microplates imaged
using a Keyence VHX System (Steinke et al. 2024b), and
larger specimens were displayed (if not already) then imaged
95 using an Automatic Imaging Rig (Steinke et al. 2024a).
Images were associated with specimen records and assigned
an Order-level identification using a machine learning algo-
rithm (Steinke et al. 2024b). For larger pinned specimens,
a small tissue fragment (typically a leg) was then removed
100 and placed into a microplate, and the remaining portion of
the specimen was labelled and placed into the Centre for
Biodiversity Genomics' Specimen Archive. Smaller specimens
are stored in 95% ethanol in microplates at room tempera-
ture in the Specimen Archive. All specimen records are linked
105 to their vouchers by unique BOLD-designated 'Sample IDs'
(Supporting information). Ultimately, specimens will be
returned to and accessioned in originating natural history
collections (Supporting information).

DNA extraction and sequencing

110 All microplates containing specimens or tissue had DNA
extracted using solid-phase reversible immobilization (SPRI)
beads. Ethanol was evaporated at 56°C, then 50 µl of lysis
buffer (1M KCl with 0.5 mg ml⁻¹ of Proteinase K) was added
115 to each well and incubated at 56°C for 18 hours. Following
incubation, 85 µl of magnetic beads (SpeedBeads Carboxylate-
Modified, GE Healthcare) was then added to each well and
incubated at room temperature for at least 10 minutes. A
120 96-pin magnetic bead extractor (407AM-N1-R, VP) then

collected the beads, and beads were washed by submersion in 150 µl of 70% ethanol. Finally, beads were air dried and DNA was eluted into 50 µl of 10 mM Tris-HCl pH 8.0 by submersion for at least 10 minutes. Following purification, all plates containing whole voucher specimens < 5 mm were deposited in the Centre for Biodiversity Genomics' Specimen Archive. Aliquots of DNA extracts from the microplates were consolidated into 384-well PCR plates for amplification, and remaining extracts were stored in a -80°C freezer archive. Unique Molecular Identifiers (UMIs) were incorporated into the amplicons to map sequence reads back to specimens, and PCR amplification (with reaction chemistry following Hebert et al. (2025) and thermocycler conditions following Hebert et al. (2018)) targeted the cytochrome c oxidase subunit I (COI) barcoding gene region (Folmer region, generally 658 basepairs). Amplicons from 24 PCR plates were pooled on a PacBio Single-Molecule Real Time (SMRT) cell for sequencing. SMRT cells were pooled prior to annealing the sequencing primer and polymerase to create the final sequencing library, which was loaded onto a SMRT cell chip and sequenced on a Sequel IIe Sequencing Platform.

Following sequencing, a high-quality consensus sequence was generated during Circular Consensus Sequencing (CCS) analysis following the approach of Hebert et al. (2025). Reads were demultiplexed using the SMRT cell adapter and UMI sequences to assign each read to its source specimen. Reads from each specimen were then clustered into contigs using VSEARCH (Rognes et al. 2016) at a 95% similarity threshold, with consensus sequences generated for contigs containing five or more reads. Contig consensus sequences were clustered a second time using VSEARCH at a 98% threshold, corresponding to the ~ 2% barcode gap that delineates many arthropod species (Hebert et al. 2009, Čandek and Kuntner 2015). Each consensus sequence was assigned to an arthropod order using VSEARCH SINTAX with a 60% minimum probability threshold, against a custom reference library of unique BOLD BINs. The full bioinformatics pipeline is described in Hebert et al. (2025; Section 2.2.5.1), with scripts available on Dryad (<https://doi.org/10.5061/dryad.41ns1rnp1>).

Taxon delimitation

The majority of our downstream analyses are based on Barcode Index Numbers (BINs), a sequence-based proxy for species. BINs were assigned to each specimen for which the consensus COI sequence was of sufficient length and quality using the BOLD Refined Single Linkage (RESL) algorithm (see Ratnasingham and Hebert (2013) for details). This algorithm first jointly estimates uncorrected pairwise distances for all sequences in the BOLD database, including both pre-existing sequences in the database and sequences generated as part of this project, and performs single linkage clustering at a distance threshold of 2.2% (Ratnasingham and Hebert 2013). The RESL algorithm finally performs Markov cluster refinement using simulated random walks to generate BINs (Ratnasingham and Hebert 2013).

A subset of our analyses made use of order-, family-, genus-, or species-level identifications. To obtain these identifications, we next associated BINs with Linnean taxonomic names using the BOLD Identification System. For each query BIN, this system translates and aligns the corresponding COI sequence using a profile Hidden Markov Model of the COI protein from the Pfam database (Eddy 1998, Finn et al. 2010, Mistry et al. 2021), calculates uncorrected pairwise distances between the query sequence and all sequences in the BOLD database, and finally assigns the query BIN to a taxon at the most specific rank at which a sufficient similarity match exists, using rank-specific similarity thresholds: 2% for species, 3% for genus, 5% for subfamily, 9% for family, and 14% for order.

In many cases, taxonomic assignments were further reviewed by expert taxonomists. These experts used information that included sequence placement in neighbour-joining trees of identified BINs and morphological assessments based on specimen images (Hebert et al. 2018, deWaard et al. 2019).

Use of Barcode Index Numbers for species delimitation and identification

We chose to use the Centre for Biodiversity Genomics' Barcode Index Numbers (BINs) to identify the specimens collected through the California Insect Barcoding Initiative. Pairing BINs with a well-developed reference library can result in rapid species-level identifications from a single inexpensive and easily sequenced gene region, and even in the absence of a reference library, BINs serve as persistent, registered operational taxonomic units suitable for biodiversity analyses. Although not a perfect proxy for species, the number of BINs estimated from large biodiversity datasets is strongly correlated with the number of morphologically identified species in those datasets (Salis et al. 2024). In the absence of comprehensive reference libraries, which do not yet exist for insects, approximating species using BINs or other molecular delimitation methods is the only practical approach for assessing patterns of biodiversity in lineages like insects that contain many dark taxa.

Data processing for BOLD and GBIF data

Specimen data from all subprojects funded by the California Insect Barcoding Initiative were obtained from BOLD on 26 June 2025. Non-hexapod data and records lacking a BIN assignment were filtered out.

In some instances, a single BIN had multiple conflicting taxonomic assignments. To address these cases, we cleaned and standardized specimen record data. If some records within a BIN had taxonomy only partially assigned (e.g. with family as the most-specific assignment) while other records of that same BIN had a more specific taxonomic identification (e.g. species as the most-specific assignment), all records for that BIN were assigned to the most specific rank possible. If records within a BIN had multiple, conflicting taxa assigned at a given rank (e.g. some records were identified as the family 'Cicadellidae' while others were identified as 'Cercopidae'), assignment was determined manually based on a combination of 1) whether either each identification had an expert

0 identifier or had been auto-assigned (see Taxon delimitation
section above; expert identifiers are indicated in Supporting
information), 2) examining the specimen images on BOLD,
and 3) comparing the number of records corresponding to
5 each taxon assignment; in most cases, there were few errone-
ous assignments mixed in with a large majority of accurate
assignments. Hereafter, we refer to BINs as species, while rec-
ognizing that BINs are operational units produced by clus-
tering of COI barcode sequences and may not correspond
10 perfectly to species delimited by other criteria. Data was also
cleaned to ensure that all latitude and longitude values for
a given collecting event were identical (removing rounding
errors) by choosing the values that were assigned to the great-
est number of records for a given event.

15 To better understand how the data generated through
the California Insect Barcoding Initiative has contributed to
our understanding of the state's insect fauna, insect records
were downloaded from the Global Biodiversity Information
Facility (GBIF) on 21 August 2025. These data comprised
20 records derived from material samples and preserved speci-
men records within California for Insecta (GBIF.org 2025a),
Collembola (GBIF.org 2025b), Diplura (GBIF.org 2025c),
and Protura (GBIF.org 2025d).

25 All sequence data, specimen images, and full specimen
records used in this study are publicly available through the
BOLD Public Data Portal (<http://portal.boldsystems.org>)
and the BOLD Application Programming Interface (API).
Individual specimen records – including sequences, photo-
30 graphs, collection metadata, and BIN assignments – can be
retrieved by searching the Sample ID listed in Supporting
information (e.g. BIOUG97379-G02). The California Insect
Barcoding Initiative's project-level data release on BOLD is
forthcoming and will allow bulk retrieval by project code.
35 Specimens for which sequencing failed do not have a pub-
licly-released specimen page on the BOLD portal but do have
Sample IDs in Supporting information.

40 **Assessing taxonomic and seasonal patterns, and
impact of collecting methods**

To assess specificity of specimen identifications, we calcu-
lated the proportion of specimens identified to the class,
order, family, subfamily, genus, and species levels. We also
45 calculated the number of species collected for each order in
our dataset. For the GBIF dataset, we calculated the number
of species per order after filtering the data for records where
the taxonomic rank was species and the taxonomic status was
accepted (Supporting information). To understand patterns
50 of species abundance, we generated rank-abundance curves
for all species in our dataset.

We assigned each collecting event and each individual
specimen to a season. Seasons were defined as follows: spring
55 (March–May), summer (June–August), fall (September–
November), and winter (December–February). We then cal-
culated the percent of collecting events and specimens for
each season.

60 Collecting methods substantially influence the taxonomic
composition of insect samples; in particular, Malaise traps are
biased towards capturing flying insects. To quantify these pat-
terns, we calculated the fraction of species that were collected
only via Malaise traps, only via non-Malaise trap methods,
65 or via both sets of methods. We also calculated the order-
level taxonomic composition for Malaise and non-Malaise
samples.

Estimating collecting bias across ecoregions

To assess the degree to which our sampling effort was even
70 across ecoregions, we calculated a collecting bias metric,
scaled such that zero indicates areas where collecting effort is
proportional to ecoregion area, as follows:

75 Collecting bias =

$$\frac{\frac{\text{Number of collecting events in the ecoregion}}{\text{Total number of collecting events}} - 1}{\frac{\text{Area of the ecoregion}}{\text{Total area of California}}}$$

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For ecoregions extending outside of California, we only con-
sidered the area of the ecoregions within California. We then
compared collecting bias for each ecoregion with the total
85 number of unique BINs collected in each ecoregion.

**Predicting insect biodiversity and estimating per-site
sampling completeness**

90 Despite significant sampling effort, it is highly unlikely that
we sampled every insect species present at each collecting site.
Therefore, we used several approaches to assess species rich-
ness and estimate a range of lower bounds on the number
of insect species present in California. First, we generated
95 rarefaction curves for 1) all samples in our dataset, and 2)
Observatory Site samples only. Second, we estimated sam-
pling completeness with the iChao1 estimator, again for 1)
all samples in our dataset, and 2) Observatory Site samples
only. Finally, we inferred a lower bound for the total number
100 of insect species based on all samples in our dataset, using the
iNEXT method from the R package 'iNEXT', which per-
forms a combination of rarefaction and extrapolation using
the Chao1 estimator.

In addition to assessing lower bounds on the total number
105 of insect species in the state, we were interested in understand-
ing how patterns of sampling completeness varied across the
landscape. To explore this, we predicted the total number of
species present at a subset of our Observatory Sites using the
iChao estimator (Chiu et al. 2014). We analyzed the subset
110 of Observatory Sites with 1) ten or more sequenced samples,
and 2) greater than 80% of taxa receiving a BIN assignment.
We then used the iChao1 estimator to obtain per-site sam-
pling completeness estimates based on the ratio of observed
115 species to predicted species at a given site.

Measuring site and ecoregion dissimilarity

We assessed whether ecoregions are useful proxies for dis-
tinct insect assemblages using both Bray–Curtis and Jaccard
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dissimilarity indices. We used Observatory Site-month combinations as the community unit in our analyses because different ecoregions had different numbers of Observatory Sites, and different Observatory Sites had different numbers of months sampled. Therefore, maintaining Observatory Site-month combinations as sampling events maintains the lowest common sampling denominator.

First, we visualized pairwise dissimilarities between ecoregions, and then we tested whether the dissimilarities of Observatory Site-month sampling events from the same, adjacent, or nonadjacent ecoregions are significantly different. For both of these, we calculated every pair-wise dissimilarity for all Observatory Site-month combinations using pairwise Bray–Curtis dissimilarity using abundance matrices (Supporting information) with the function *vegdist* from the R package ‘vegan’ (Oksanen et al. 2013). Similarly, we calculated pairwise Jaccard dissimilarity indices using presence/absence matrices (Supporting information) with the function *beta.pair* from the R package ‘betapart’ (Baselga and Orme 2012). Using the *beta.pair* function, we also partition the contributions of nestedness and turnover to the Jaccard beta diversity for every pair of site-month sampling events.

For the visualization, we simply averaged each of these metrics (beta diversity using Bray–Curtis, Jaccard, Nestedness, and Turnover) across every pair of EPA Level III ecoregions.

Then, to further explore patterns of community dissimilarity across the landscape, we tested whether Observatory Sites located in the same EPA Level III ecoregion have more similar insect communities than Observatory Sites located in either adjacent or nonadjacent ecoregions. We classified pairs of ecoregions as adjacent or nonadjacent using the function *st_intersects* from the R package ‘sf’ (Pebesma and Bivand 2023). We then utilized Hierarchical Bayesian zero/one inflated beta regressions (for each of the metrics discussed above) to assess differences in mean pairwise dissimilarities across the three categories of site pairs (same ecoregion, adjacent ecoregions, nonadjacent ecoregions) while accounting for site identity and month of sampling. For these analyses, Observatory Site Malaise samples were used as replicates, and site identity and sampling month were included as random factors. Because pairwise dissimilarity values will not be independent when comparing two samples from the same Observatory Site or two samples collected in the same month, we used multi-membership random effects, which account for the fact that a sample is drawn from a pair of sites in a pair of months. This random effect then accounts for the non-independence of these comparisons. Multimembership random effects are equivalent to site-block bootstrap in a Bayesian analysis but allow for multiple multi-membership effects, such as those we have here (i.e. both site membership and month membership). Models were run specifying a zero/one inflated Beta regression, with 4 chains, 7 cores, 20 000 iterations, and a maximum tree depth of 20 in the ‘brms’ package using RStan (Bürkner 2017).

Finally, to complement this analysis, we performed an NMDS plot to visualize differences in community

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dissimilarity across ecoregions, using the Jaccard index and three axes to maintain stress between 0.1 and 0.2.

Inferring the potential distribution of insect biodiversity across California

We inferred potential distributions by projecting observed occurrences onto combinations of ecoregions and vegetation types where taxa were sampled, generating habitat-constrained range estimates at 1-km resolution. For each species, rule-based spatial range maps were generated following a modified version of the approach used by (Pinkert et al. 2023) and (Chesshire et al. 2023). We chose this approach over the minimum convex hull approach, because the latter method can only produce contiguous range maps, thus inflating the predicted biodiversity of Central California where most species ranges would artificially appear to overlap. Additionally, our approach permits the inclusion of species represented by as few as a single specimen because it does not rely on drawing polygons around multiple specimen records.

To generate these range maps, we associated each species with 1) the Level IV ecoregions where it occurred in our dataset, using EPA ecoregion shapefiles (U.S. Environmental Protection Agency 2024), and 2) the vegetation types where specimens were sampled, using Existing Vegetation Type (EVT) data accessed from the LANDFIRE database (LANDFIRE 2022). Ecoregional maps were then filtered to only contain the LANDFIRE vegetation types where species were found. The EVT data are at a 30 m resolution and our data had over 42 000 species; therefore, to improve computational speed, species range maps were inferred at a 1 km resolution across California.

To understand aggregate patterns of sampled species richness across the state, we hierarchically combined the individual range maps into family-level, order-level, and all taxa maps (Supporting information). These maps display the potential estimated species richness across California at a one-kilometer resolution. Unlike traditional species distribution models, which statistically estimate environmental response functions, our approach uses observed occurrences combined with ecological stratification (ecoregion + vegetation type) to interpolate presence across unsampled locations. This method is scalable to tens of thousands of species without requiring environmental covariates or parametric inference.

Results

Taxonomic patterns

The California Insect Barcoding Initiative project has attempted sequencing for 1 030 728 insect specimens as of 25 June 2025. Of these, 892 274 specimens yielded BIN-compliant (Ratnasingham and Hebert 2013) barcode sequences, comprising 42 305 unique BINs (used here as a proxy for species; see Methods) (Supporting information).

The specificity of taxonomic identification (rank) for specimens with BINs is mostly at the family (38.4%), genus (24.2%), or species levels (26.5%), while fewer specimens have been identified only to order (1.7%) or class (< 1.0%) (Fig. 1A). Additional specimens from this project continue to be submitted for sequencing and BIN assignment, and identifications will improve as additional reference specimens are sequenced and added to the BOLD barcode library.

In our sampling, the orders of insects with the highest levels of diversity are Diptera (16 181 species, comprising 38.2% of all sampled species), Hymenoptera (13 966 species, 33.0%), Coleoptera (4234 species, 10.0%), Lepidoptera (3126 species, 7.3%), and Hemiptera (2934 species, 6.9%), with 1864 species for all other orders combined (4.4%) (Fig. 1B). There are substantial differences between our dataset and the insect records for California present in the Global Biodiversity Information Facility (GBIF) database. The GBIF records comprise 17 854 species belonging to the following orders: Coleoptera (5186 species; 29.1% of all species), Hymenoptera (3611; 20.2%), Lepidoptera (3520; 19.7%), Diptera (2353; 13.2%), Hemiptera (1568; 8.8%), and all others combined (1616; 9.1%) (Fig. 1B).

The vast majority of species sequenced could be considered uncommon, given the low frequency with which they occur in our data, with 78.7% of species represented by fewer than 10 sequenced individuals (Fig. 1C). On the other end of the spectrum, 103 species were so abundant that 1000 or more individuals were sequenced from our samples. One species is represented by 14 841 individuals, and corresponds to a super-abundant species of grass fly (Chloropidae: Oscinellinae: *Olcella* sp.). This species was collected year-round and detected in 11 out of 13 level III ecoregions across the state; we failed to collect it only in the far northeastern corner of the state, where sampling effort is lowest. We do

note that specimens, particularly those from Snapshot Sites, were sorted prior to sequencing to reduce redundancy; this would particularly impact abundance estimates for eusocial insects (honeybees and ants) and collembolans.

Impact of collecting method

The composition of sampled specimens is likely shaped by our sampling methods, which were dominated by Malaise traps. For samples sequenced as part of this dataset, our Malaise traps ran for over 170 000 trap hours, and this collecting effort produced 91.6% of all sequenced specimens. All other collecting techniques are responsible for just 8.4% of the total number of specimens (primarily vegetation sweeping, 5.0% of all specimens). However, these collecting techniques captured a disproportionately large fraction of unique species: 13.8% of species were collected only via non-Malaise methods, while 64.4% of species were collected only by Malaise traps, and 20.8% of species were collected using both approaches (Supporting information).

The specimens collected via non-Malaise trap methods display a different taxonomic composition than do specimens collected in Malaise traps. Of the most abundant orders, Hemiptera make up a much higher proportion of non-Malaise trap specimens than Malaise trap specimens (19.9% versus 7.1%), as do Coleoptera (10.2% versus 4.0%) and Hymenoptera (20.0% versus 14.9%) (Supporting information). Nearly all minor orders were more abundant among non-Malaise trap specimens. The non-Malaise methods also yielded specimens for two flightless groups that were never collected in Malaise traps, Acerentomata and Siphonaptera.

Seasonal patterns in collecting

The seasonal distribution of collecting events was fairly even across spring (March–May: 29.9%), summer (June–August:

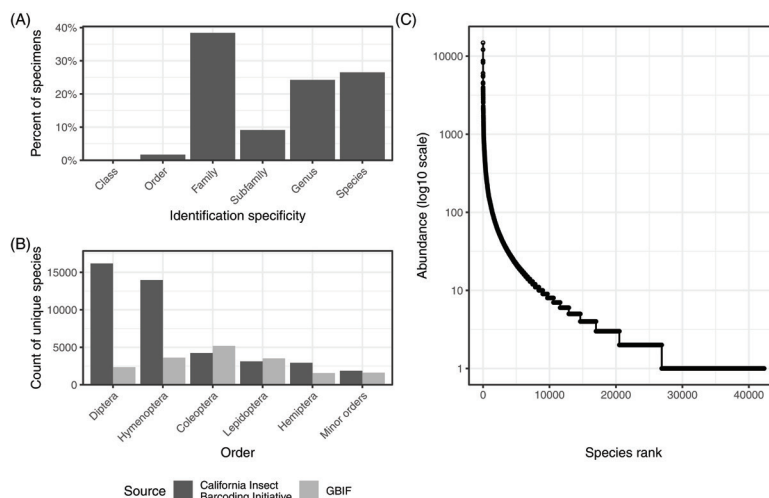


Figure 1. Taxonomic and abundance statistics for California Insect Barcoding Initiative data. (A) the taxonomic specificity of identifications assigned to specimens. (B) for our dataset (dark gray) and for California insect data derived from GBIF (light grey), the number of unique species for each of the 5 most abundant orders of insects and for the minor orders in aggregate (those with fewer than 2400 species in our dataset). (C) rank–abundance curve for species identified in this project; the y-axis presents the number of individual specimens collected for each species, and the x-axis presents the ordinal rank for each species from most to least abundant.

37.3%), and fall (September–November: 27.9%), but lower in winter (December–February: 4.8%). In contrast, the seasonal distribution of specimens sequenced is biased towards the fall (63.2%) relative to spring (17.1%), summer (19.1%), and winter (0.5%), with a larger proportion of sequenced specimens being caught during fall in the Observatory Site Malaise traps.

Collecting biases across ecoregions

We sampled widely across California, sampling all level III regions and 80% of level IV regions by the end of 2024 (Fig. 2A). Ecoregions varied in the number of species collected and sequenced, from 10 585 in the Southern California Mountains surrounding Los Angeles and San Diego, to 625 in the Northern Basin and Range (Fig. 2B). While this variation could reflect true differences in species richness, it may also reflect biases in collecting effort. To assess these potential biases, we developed a metric that measures collection effort relative to the area of each ecoregion (Fig. 2C). In this way we show that the Southern California/Northern Baja Coast is relatively over-collected, likely because three major institutional participants (NHMLAC, UCR, and SDNHM) are

present in this region. In contrast, the Central California Valley, for example, is relatively under-collected. This bias likely results from intense agricultural development in the Central Valley. As a result, the majority of the land tenure is private, making the Central Valley the most difficult ecoregion in which to find natural areas to sample, despite its large size.

Predicted biodiversity and per-site sampling completeness

Based on the asymptote obtained using rarefaction/extrapolation with the Chao1 estimator, the estimated lower bound for the number of species present in California is approximately 60 945 (CI: 60 339–61 551). Using the iChao1 estimator for all sequenced specimens, the estimated lower bound given our current sampling is approximately 64 915 species. In contrast, using the iChao estimator with only Observatory Site specimens produces an estimated lower bound of approximately 49 535 Malaise-trappable species in California, given our current sampling. Neither the species accumulation curves nor the rarefaction curves yet reach asymptotes (Fig. 3A).

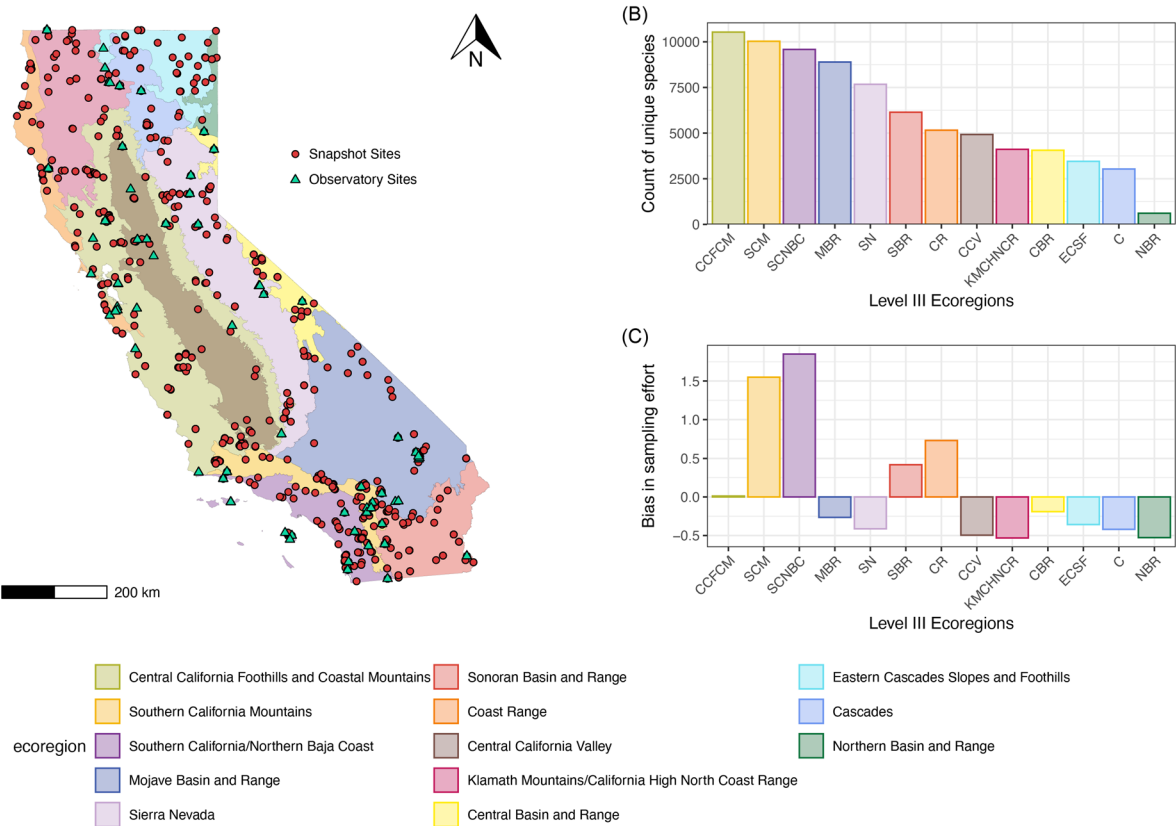


Figure 2. Collecting effort across California. (A) distribution of sampling events across California’s EPA Level III ecoregions, with blue triangles indicating Observatory Sites and red circles indicating Snapshot Sites. (B) the total number of unique species collected per ecoregion. (C) the relative collection effort per ecoregion corrected by the relative area of each ecoregion, scaled such that values of zero indicate areas where collecting effort is equal to ecoregion area, values greater than zero represent areas that are over-collected, and values below zero represent areas that are under-collected; values = (EC/TC)/(EA/TA) - 1; EC = ecoregion collecting events; TC = total collecting events; EA = ecoregion area; TA = total area.

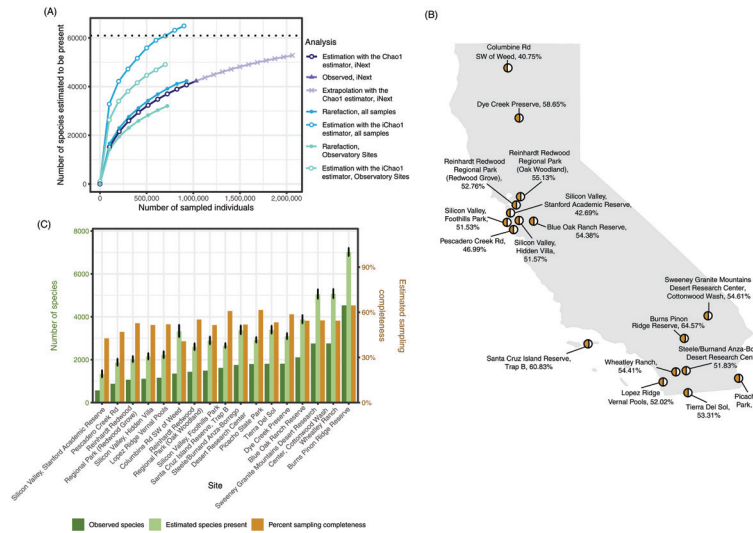


Figure 3. Species richness and sampling completeness estimates. (A) rarefaction and species estimate curves for three methods used to estimate statewide sampling completeness and a lower bound for the number of species present in the state of California. Analysis of all samples using the Chao1 estimator implemented in iNext are presented in purple; analysis of all samples using rarefaction or estimation with the iChao1 estimator presented in blue; and analysis of Observatory Site samples using rarefaction or estimation with the iChao1 estimator presented in teal. The dashed line indicates the asymptote value inferred using the Chao1 estimator rarefaction/extrapolation analysis. (B) approximate locations of the 17 best-sampled Observatory Sites, with estimated percent sampling completeness presented as moon plots. (C) counts for the actual number of unique observed species (dark green bars), estimated number of species present (light green bars), and percent sampling completeness (orange bars) for each of the 17 sites.

We also inferred per-site sampling completeness for the 17 best-sampled Observatory Sites (Fig. 3B–C). Estimated completeness ranged from 40.7 to 64.6%, with a mean of 53.4%. Because our samples were largely collected via Malaise traps, which undersample non-flying insects, we note that these diversity metrics are underestimates of total insect species richness at these sites.

Site and ecoregion dissimilarity

Our data indicate that California's EPA level III ecoregions host highly dissimilar insect communities (Fig. 4A), with a mean pairwise Bray–Curtis dissimilarity index of 0.94

(range: 0.92–0.99). These high values are consistent with the extreme rarity structure and turnover characteristic of hyperdiverse insect assemblages. In the context of this high between-ecoregion dissimilarity, the Klamath Mountains/High North Coast Range and the Cascades are more similar to one another, suggesting a somewhat shared insect fauna of the mountains of Northern California. The Sonoran Basin and Range and the Mojave Basin and Range are also more similar, indicating that these desert ecoregions share somewhat similar insect communities. Nevertheless, we emphasize that all ecoregions host highly dissimilar insect communities.

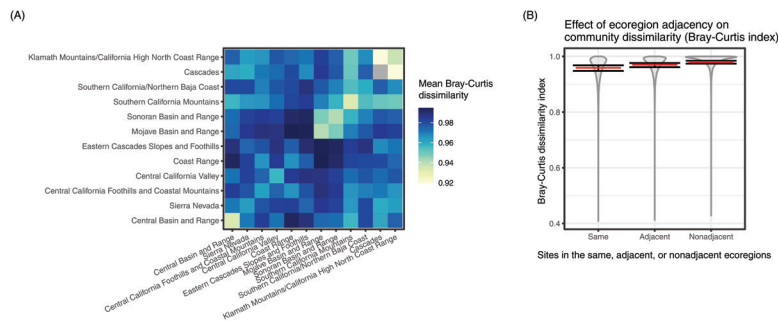


Figure 4. Insect community dissimilarity across ecoregions and Observatory Sites. (A) heatmap of average site pairwise Bray–Curtis dissimilarity index for each pair of California EPA level III ecoregions (excluding the Northern Basin and Range, where we do not have any Observatory Site locations). Darker blue colors, corresponding to higher values of the Bray–Curtis dissimilarity index, indicate less similar insect communities, while lighter green and yellow colors indicate more similar insect communities. (B) pairwise Bray–Curtis dissimilarity index for sites within the same ecoregion, within adjacent ecoregions, and within nonadjacent ecoregions; red bars indicate the estimated mean value from a Bayesian zero/one inflated regression for each site pair category, and black crossbars indicate the corresponding credible interval.

In general, Observatory Sites within the same ecoregion are more similar to one another than are sites in adjacent ecoregions, and sites in adjacent ecoregions are in turn more similar than sites in nonadjacent ecoregions (Fig. 4B). Observatory Sites within the same ecoregion have a mean estimated Bray–Curtis dissimilarity of 0.96 (SE=0.13; 0.95, 0.97 credible interval). Observatory Sites within adjacent ecoregions have a mean estimated Bray–Curtis dissimilarity of 0.97 (SE=0.01; 0.96, 0.98 credible interval). Observatory Sites within nonadjacent ecoregions have a mean estimated Bray–Curtis dissimilarity of 0.98 (SE=0.01; 0.97, 0.98 credible interval). While we reported the means and credible intervals in terms of the response scale (back-transformed), within the linear regression, the difference between the intercept (same category), and both the adjacent (mean difference=0.31, 0.01, 0.3, 0.33) and nonadjacent (mean difference=0.73, 0.01, 0.71, 0.75) was significantly different from zero.

The standard deviation captured by the combination of month and site random effects is 0.82 (0.71, 0.96 credible interval), indicating that a meaningful amount of variation in dissimilarity is explained by site identity and month of sampling. We find that turnover contributes to most of the beta diversity observed, compared to nestedness (Supporting information). Not surprisingly, we found that as site-months are compared within the same ecoregion, across adjacent ecoregions, and across nonadjacent ecoregions, beta diversity not only increases, but the contribution of turnover also increases (Fig. 4, Supporting information). Nonetheless, we find that even within the same ecoregion, nestedness does not contribute much to beta diversity. This makes sense for hyperdiverse groups with small ranges. Overall, we find no differences between the Jaccard index analyses and our analyses done with Bray–Curtis indices (Fig. 4, Supporting information). All R-hat values are equal to 1.

Further, the NMDS analysis showed consistent results to the Bayesian analysis (Supporting information). Across NMDS axes 1 and 2, and 1 and 3, we find gradients of community composition, where ecoregions segregate along axis 1. Ecoregions that load highest or lowest on axis 1 do not overlap, suggesting high dissimilarity in community compositions among ecoregions. The ecoregions that load in the center of axis 1 overlap substantially in community composition, and similar to our Bayesian analysis, we find that adjacent ecoregions overlap more in the NMDS plot. Similarly, ecoregions that are geographically central, such as the Central California Valley, show up at the center of the NMDS plot. Finally, we find that dissimilarity among sampling events is high, as shown by the large size of the ellipses.

The spatial distribution of insect biodiversity in California

Our potential biodiversity heatmap (Fig. 5) identifies areas of particularly high insect species richness in several Level IV ecoregions within the Southern California Mountains, including the Arid Montane Foothills, Southern California Montane Conifer Forest, and Southern California Lower Montane Shrub and Woodland. Each of these regions

contains one-square-kilometer areas that are predicted to contain over 8000 insect species. Additionally, the Mojave Basin and Range contains level IV ecoregions with high predicted diversity, including the Eastern Mojave Low Ranges and Arid Foothills and the Eastern Mojave Basins (also predicted to have over 8000 insect species).

Our diversity predictions only take into account sampled species, and are therefore influenced by sampling effort. Thus, while we infer high species richness in the Southern California Mountains, this ecoregion is also highly over-collected in our dataset (Fig. 2C). However, the Mojave Basin and Range, another area of high inferred insect diversity, is actually undercollected, suggesting that this region contains true peaks in diversity.

The effect of sampling effort can also be seen at smaller spatial scales. For example, the Burns Piñon Ridge Reserve, which includes one of the most heavily sampled Observatory Sites, has a predicted richness value of 7092 species based on species collected both at that site and other sites with the same vegetation types and in the same ecoregion. Given that our predicted sampling completeness for the Observatory Site Malaise trap at that location is 64.57%, this result suggests that there may be as many as 10 983 insect species at that location.

Our interpolated diversity maps also highlight remaining gaps in sampling, corresponding to vegetation type-ecoregion combinations from which we do not yet have sequenced specimens. Some of these areas, indicated in gray, have been sampled but specimens have not yet been sequenced and therefore cannot be included in the present analysis. Many sampling gaps fall in highly agriculturized Central Valley ecoregions, where collecting is challenging.

The patterns that we observe for all insects are largely mirrored at the order level, particularly for groups with sufficient sampling and species diversity to yield informative richness indices (i.e. Coleoptera, Diptera, Hemiptera, Hymenoptera, Lepidoptera, Neuroptera, Orthoptera, and Thysanoptera) (Supporting information).

Discussion

The California Insect Barcoding Initiative represents one of the largest efforts ever made to document regional insect biodiversity using DNA barcoding, with 1 030 728 hexapod specimens processed to date. We maximized the biodiversity collected by sampling in every EPA Level III ecoregion across California. This effort resulted in 42 305 unique BINs collected and sequenced, used here as a proxy for species (Ratnasingham and Hebert 2013). Using these data, we infer statewide patterns of insect biodiversity and demonstrate that California's ecoregions harbor strongly dissimilar insect communities, with overall diversity peaking in the Southern California Mountains ecoregion and the Mojave Basin and Range ecoregion. The barcode sequences generated by this project also represent a major contribution to the development of a reference library that can be used for future rapid

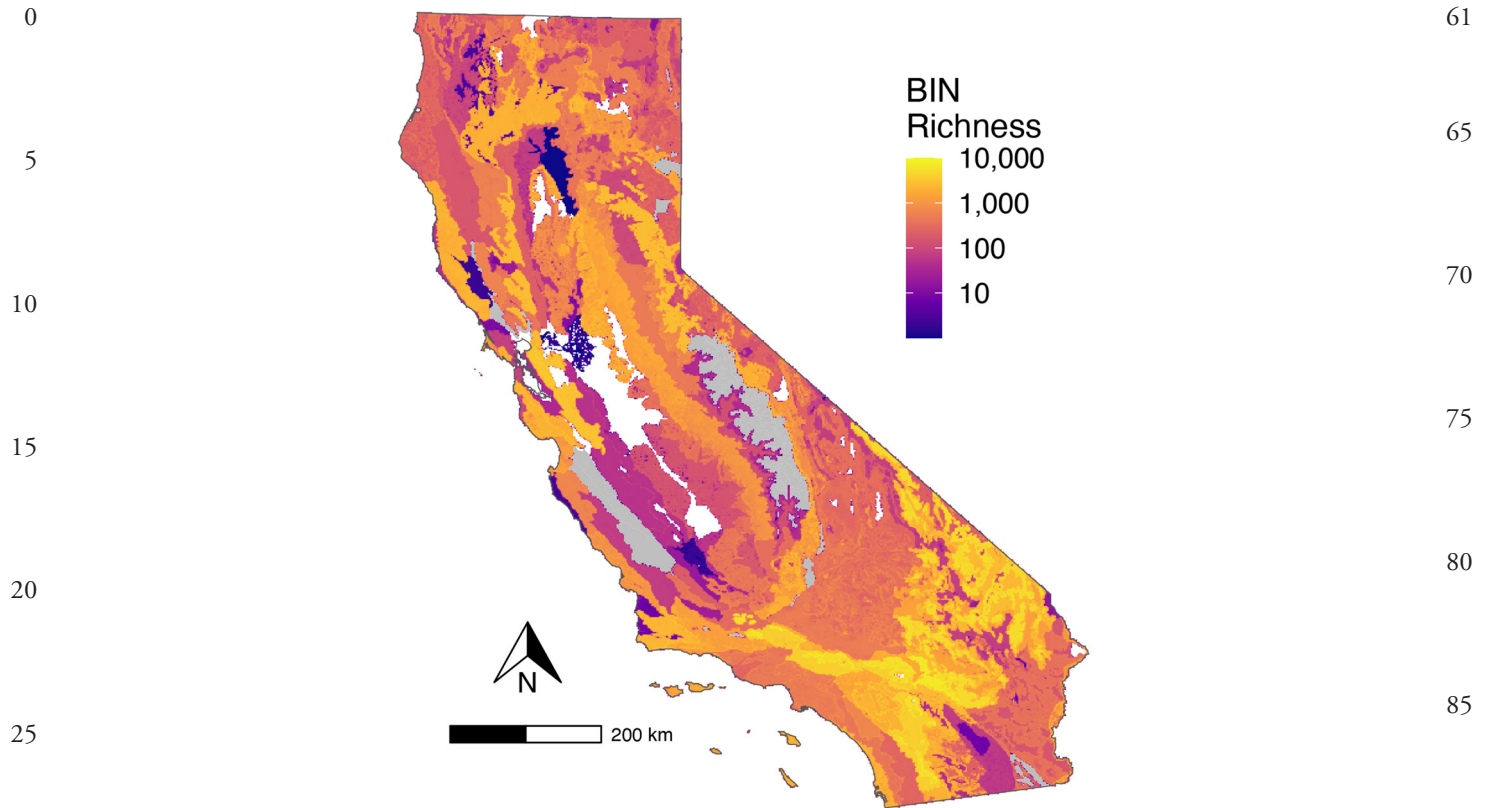


Figure 5. Geographic distribution of potential richness estimate for insects across California. Estimates are based on rule-based spatial range interpolation constrained by ecoregion and vegetation type for every species in the dataset. The number of species predicted in each square kilometer is represented by the color plotted on the map, with lighter colors indicating higher species richness (color scale log-transformed), gray indicating EPA level IV ecoregions that have been sampled but not yet sequenced, and white indicating ecoregions and EVT cover type combinations for which we lack insect samples.

and affordable species identifications. Here we first evaluate sampling completeness, then assess spatial turnover across ecoregions, and finally consider implications for conservation planning and future monitoring.

Collecting methodology strongly impacts the results of insect surveys, as different methods preferentially target different taxa (Anderson et al. 2024, Souto-Vilarós et al. 2025). Like other large-scale insect surveys (Hebert et al. 2016, Karlsson et al. 2020a, b), the California Insect Barcoding Initiative heavily relied on Malaise trapping. While Malaise traps are easily deployed and allow consistent passive collection, they do not effectively sample groups that do not fly frequently or at all (e.g. many beetles, springtails, aquatic insects, scale insects) or insects that are able to see and avoid traps (e.g. dragonflies, butterflies, etc.) (Lamarre et al. 2012, Devigne and De Biseau 2014, Montgomery et al. 2021, Srivathsan et al. 2023, Souto-Vilarós et al. 2025). Our extensive use of Malaise traps did significantly influence the composition of collected specimens. However, to mitigate these sampling biases, we employed a variety of additional collecting methods which increased the diversity of insects collected. This approach has generated a highly biodiverse dataset, but one that is also more heterogeneous compared to those that

target a single community or use a single collecting method (Caterino and Recuero 2024).

Our study also demonstrates the utility of using DNA barcodes to understand insect biodiversity at large scales. Compared to projects that rely on morphological identification, such as the Swedish Malaise Trap Project (Karlsson et al. 2020a, b), the California Insect Barcoding Initiative was able to rapidly sequence, database, and delimit over 890 000 BIN-compliant specimens in three years. Barcoding is also advantageous in that it permits the identification of cryptic species. Since cryptic species appear to comprise a substantial fraction of total insect biodiversity (Li and Wiens 2023, Basset et al. 2025), it is critical that surveys of insect fauna utilize species identification approaches that are capable of detecting these species. The main disadvantage of barcoding is that specimens are initially delimited to BINs rather than named species; currently only 26.5% of specimens have been assigned species names. However, ongoing efforts to generate barcodes from authoritatively identified specimens that are vouchered in legacy collections will facilitate more precise, traditional species identifications, as will additional rigorous taxonomic work.

California is tremendously biodiverse (Myers et al. 2000, Calsbeek et al. 2003), and despite decades of effort we are

just now beginning to quantify the state's insect diversity. The California Insect Barcoding Initiative builds upon previous statewide surveys led by the University of California in the 1940–1980s and between 2015 and 2020. These efforts collectively described approximately 12% of California's insect diversity, resulting in estimates that there are 26 000–100 000 insect species present in the state, of which 6% were considered new to science (University of California, Davis 2022).

Estimates of the number of insect species present in the state can also be produced by leveraging digitally-available legacy specimens. For instance, the GBIF database contains 17 854 insect species represented by specimens collected in California, less than half the number of species (BINs) collected and sequenced by the California Insect Barcoding Initiative. Discrepancies are particularly pronounced in some groups: for instance, our dataset contains 688% more Diptera species and 387% more Hymenoptera species for California than does GBIF. These differences likely reflect both our use of DNA barcoding for species delimitation, allowing us to more easily make taxonomic identifications, detect cryptic lineages, and detect and distinguish previously undescribed species (Meier et al. 2025) and the limited proportion of North American collections holdings that are currently digitized (ca 5% of pinned insect specimens) and thus discoverable via GBIF (Cobb et al. 2019).

Despite several years of significant sampling effort statewide, rarefaction curves demonstrate that we have not yet sampled every insect species found in California. We therefore estimate the likely number of insect species present in the state, inferring a conservative lower bound of approximately 50 000 to 61 000 species in California. Our analyses further indicate that the project's sampling completeness is 65.2–69.4%. We note that we used different methods and subsets of the data to estimate these lower bounds because each method and subset of the data has different strengths and weaknesses. First, because the iChao1 estimator is sensitive to the number of uncommon species (Chiu et al. 2014), we expect that implementing this estimator for the Observatory and Snapshot Sites in combination will produce an overestimate of species richness, as targeted collecting and sorting of Snapshot Site samples results in an increased number of rare species. However, implementing it for Observatory Sites alone will produce an underestimate, as species captured only via Snapshot Site sampling will be excluded. Second, relative to the iChao1 estimator, the Chao1 estimator underestimates species richness (Chiu et al. 2014); however, using this estimator as implemented in 'iNext' allowed us to extrapolate beyond our current sampling to obtain an estimated lower bound for the total number of species present in California (Hsieh et al. 2016). While our extrapolation methods are best suited to collecting methods that are not biased towards rare or abundant taxa (Chao et al. 2009), we decided to present the most inclusive estimate to provide a more comprehensive view of the potential insect diversity in California. Finally, while 61 000 species is a large number, we expect this is potentially an underestimate since it is heavily based on

sampling done with Malaise traps, which only comprehensively sample some insect groups (Souto-Vilarós et al. 2025).

Given these caveats, the true number of species present in California is likely to be well above the lower bound of 61 000, although the degree to which our current prediction is an underestimate is difficult to ascertain. Nevertheless, it is clear that our sampling has not yet fully captured the total number of insects present in the state. This pattern is not atypical for insect biodiversity studies. For example, in an ongoing long-term insect survey of Costa Rica's Area de Conservación Guanacaste (Quicke et al. 2024), species accumulation curves have not yet reached an asymptote. This begs the question: if more than 40 years of continuous sampling have been insufficient to collect every insect species in a 147 000 hectare area of Costa Rica, how much sampling will be needed for the 42 million hectare state of California?

The California Insect Barcoding Initiative also sheds light on the potential spatial distribution of insect biodiversity. Across the state, we find that insect species richness appears to peak in the Southern California Mountains and in the Mojave Basin and Range ecoregions. Biases in collecting effort may have inflated biodiversity estimates in the Southern California Mountains. However, the Mojave Basin and Range is actually under-sampled relative to the area of the ecoregion, and thus likely represents a true peak in insect biodiversity.

On the whole, we find that sites across the state host strongly dissimilar insect communities. Community dissimilarity is high both within and between ecoregions, although ecoregions in closer spatial proximity are slightly more similar. Interestingly, two ecoregions which encompass Northern California mountain ranges, the Cascades ecoregion and the Klamath Mountains/California High North Coast Range ecoregion, are the two most similar ecoregions in terms of their insect fauna. These two ecoregions share many similarities in tree diversity, elevation, and climate (Griffith et al. 2016), in addition to being adjacent to one another. Likewise, the state's two desert ecoregions, the Mojave Basin and Range and the Sonoran Basin and Range, are more similar to one another than to other ecoregions and are highly dissimilar to all other ecoregions in the state. This pattern suggests that the warm, arid, and strongly seasonal abiotic conditions in these deserts drive common patterns of insect community composition.

To date, our understanding of the spatial distribution of biodiversity across California has largely been based on five groups of organisms: birds, mammals, amphibians, reptiles, and plants (California Department of Fish and Wildlife 2018). Data for these groups are compiled in the California Department of Fish and Wildlife's Areas of Conservation Emphasis Terrestrial Native Species Richness maps, which are used to identify biodiversity hotspots and areas for future conservation under the state's 30 × 30 initiative. However, none of these taxa, individually or in aggregate, exhibit the same pattern of diversity as the insects. For instance, in some areas of the Mojave Basin and Range where we find high

values of insect diversity, the ACE Terrestrial Native Species Richness Summary reports diversity values that fall in the lowest quintile statewide. Thus, because these five groups appear to be imperfect proxies for insects (and likely other invertebrates), a failure to explicitly consider insects in spatial conservation planning will result in a failure to protect this important component of terrestrial biodiversity.

The data generated via the California Insect Barcoding Initiative have substantial applications beyond this study. As discussed above, they have important implications for state and regional conservation planning. Moreover, our efforts are contributing to the generation of a robust DNA barcode reference library for insects in the state of California; while at the moment this reference library is primarily useful to identify BINs, as only 26.5% of the BINs are identified to species, this library builds a foundation for the future as species-level identifications improve. This reference library will facilitate future investigation of insect diversity in the region; in particular, it has the potential to substantially expand the reference framework available for environmental DNA (eDNA) survey efforts, wherein DNA present in environmental samples is collected and sequenced in lieu of directly sampling target organisms. Successful taxonomic identifications in eDNA metabarcoding analyses depend on the availability of reference sequences, which currently have poor taxonomic resolution for most insect groups (Leandro et al. 2024). Thus, by greatly increasing the number of reference sequences available for California insects, the California Insect Barcoding Initiative will improve the accuracy of future eDNA surveys in the region.

Moreover, the abundance and diversity data collected by the Observatory Sites permit phenological inferences and can be used as a starting point for long term population or community monitoring. The exact location and time at which each standardized Malaise trap was deployed is recorded, and many Observatory Sites are paired with weather stations from which abiotic data can be derived. Moreover, the wet-weight biomass and taxonomic composition of the Observatory Site samples can be directly compared to future replicate samples to understand changes in insect communities over time. In the face of current insect declines and the increased anthropogenic movement of species, establishing insect diversity, abundance, and distribution baselines now is more important than ever.

In addition to producing a resource to enable future biodiversity studies, this dataset has great utility for detecting both native species and species not previously known from California and can be used as a starting point for tracking the distributions and movement of invasive species. For example, we detected an Asian aphid species, *Shinjia orientalis*; an aphid species native to the eastern United States, *Hyperomyzus picridis*; and a Floridian fire ant, *Solenopsis abdita*. None of these species were previously known to occur in California. For several minor orders, we actually find that the most abundant species is introduced. For example, the European earwig, *Forficula auricularia*, is by far the most abundant dermapteran; the black webspinner *Oligotoma*

nigra, which is originally from India, is the most abundant embiopteran; and the cosmopolitan dimorphic springtail, *Entomobrya atrocincta*, is the most abundant entomobryomorph Collembole.

The California Insect Barcoding Initiative has made significant progress towards creating a comprehensive DNA barcode reference library for the insect fauna of California, but we have a long way to go before this goal is fully met. For instance, the current dataset contains sampling gaps, both taxonomic and geographic. These gaps continue to be filled by additional targeted collecting throughout 2025. However, as sampling progresses, an increasingly large fraction of the remaining unsampled species will be rare taxa. These species will likely require large amounts of collecting and/or the use more labor-intensive collecting methods to sample.

In addition, fully identifying sequenced specimens to named species, rather than to BINs, will be a substantial and ongoing challenge. Barcode sequencing of expertly-identified specimens from legacy collections will help ameliorate this challenge; however, the large number of undescribed and cryptic species (BINs) that have been discovered as part of this survey cannot be identified in this manner. Large scale DNA barcoding datasets, such as the one generated by our project, may be used to accelerate the formal description of taxa, for instance by diagnosing particular groups of dark taxa, allowing taxonomists to prioritize these groups. To increase the efficiency of taxonomic discovery, we must understand where the largest gaps in our taxonomic knowledge are before we lose unknown species to extinction. If those specimens are to ever have species identifications, then we must prioritize investing resources into the advancement of basic taxonomic revisionary science. These next steps will be difficult but will greatly improve our understanding of California's diverse insect fauna.

Conclusion

The California Insect Barcoding Initiative represents one of the most extensive DNA barcoding efforts yet undertaken for regional insect biodiversity and demonstrates that high-throughput sequencing can transform our understanding of macroecological patterns. By generating more than one million specimen records, we show that insect richness in California is far greater than previously documented and that spatial patterns of diversity emerge which have not been captured in broad-scale biodiversity assessments of vertebrate and plant taxa.

Beyond establishing a foundation for a scalable reference library, our results underscore the power of DNA barcoding to detect cryptic diversity, reveal novel biogeographic patterns, and provide baselines for monitoring community change in the face of global insect declines. While taxonomic resolution and sampling completeness remain challenges, these limitations highlight the urgent need to integrate molecular inventories with revisionary taxonomy and targeted sampling. Continued development of this and similar resources will not only refine estimates of species richness and distributions, but

0 also contribute to understanding the drivers of insect diversity and to strengthening the role of invertebrates in conservation planning.

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